中国海洋大学海德学院

HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

Statistical Modelling and Inference

Week 9

Outline:

i. Estimation of parameters

ii. Properties of residual's

Theorem's:

Suppose that $Y_1, Y_2, \dots, Y_n$ are independent with $E(Y_i) = \beta_0 + \beta_1 X_i$ and $Var(Y_i) = \sigma^2$, then show that:

i. $E(\hat{\beta}_0) = \beta_0$ and $E(\hat{\beta}_1) = \beta_1$

ii. $var(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}}\right)$

iii. $var(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$

iv. $\mathrm{cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{X}}{S_{xx}}$

Theorem's:

Suppose that $Y_1, Y_2, \ldots, Y_n$ are independent with $E(Y_i) = \beta_0 + \beta_1 X_i$ and $Var(Y_i) = \sigma^2$

Then show that:

i. $E\left(\hat{\beta}_0 + \hat{\beta}_1 X_0\right) = \beta_0 + \beta_1 X_0$

ii. $Var\left(\hat{\beta}_0 + \hat{\beta}_1 X_0\right) = \sigma^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{S_{xx}}\right)$

Where:

$$E\left(\hat{\beta}_0\right) = \beta_0 \,,\ E\left(\hat{\beta}_1\right) = \beta_1 \,,\ Var\left(\hat{\beta}_0\right) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}}\right)$$

$$Var\left(\hat{\beta}_1\right) = \frac{\sigma^2}{S_{xx}} \qquad \text{and} \qquad \mathrm{Cov}\left(\hat{\beta}_0, \hat{\beta}_1\right) = -\frac{\sigma^2 \bar{X}}{S_{xx}}$$

Properties of Residuals (Error's)

Properties of residual $\hat{e}_i = Y_i - \hat{Y}_i$

i. $\sum_{i=1}^{n} \hat{e}_i = 0$

ii. $\sum_{i=1}^{n} \hat{e}_i X_i = 0$

iii. $E(\hat{e}_i) = 0$

iv. $Var(\hat{e}_i) = \sigma^2 \left(1 - \frac{1}{n} - \frac{(X_i - \bar{X})^2}{S_{xx}}\right)$

Estimation of σ^2

To estimate σ^2, we use the residual variance

$$S_e^2 = \frac{1}{n-2}\left[S_{yy} - \hat{\beta}_1 S_{xy}\right]$$

Where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ is the fitted and predicted value for the observation.

Thank

You

谢谢